

Study Guide for Test #1

Note: This study guide is meant to help you pinpoint specific topics that will be on the upcoming test. All information covered in lecture and assigned readings is fair game for the test regardless of whether it is mentioned specifically on this study guide.

Studying the Nervous system

What is the nervous system?

What are the different branches of the nervous system?

Give examples of processes associated with activation of each part of the nervous system.

Be familiar with the makeup of the spinal nerves.

What are the names of the divisions of the spinal cord?

How many nerves are included in each division?

What information comes into the nerve? What information is sent out?

Be familiar with the cranial nerves by name, number, and sensory/motor function

What are the different types of glial cells?

What are the primary functions of each type of glial cell?

What are the different parts of a neuron?

What is the typical function of each part of the neuron?

What are the differences between axons and dendrites?

What are the factors by which neurons are typically classified?

What parts of the body protect and support the brain?

How are they organized going from outside the body to closest to the brain?

What is cerebrospinal fluid?

What structures distribute CSF throughout the brain?

What structure connects the left and right cerebral hemispheres?

What is the structure made of?

What are the structures of the forebrain?

What are the subdivisions of the major structures?

What kind of behaviors and functions are associated with activation of the specific structures?

What are the structures of the midbrain?

What are the subdivisions of the major structures?

What kind of behaviors and functions are associated with activation of the specific structures?

What are the structures of the hindbrain?

What kind of behaviors and functions are associated with activation of the structures?

Think about activities you might do in a day and which brain structures are likely to be active while you do a specific activity.

Electrical Signals

Be able to describe the series of events involved in neural transmission from resting potential to synaptic transmission

What are the different parts of the cellular membrane?

How does each part contribute to permeability of the cellular membrane?

What are the main ions that are important players in neural transmission?

How do these ions move through the cell membrane?

What are the two forces that influence the movement of ions and how does each force work?

What does it mean for an ion to reach equilibrium?

What does the internal and external environment of a neuron look like when the cell is at Rest (ion concentrations and net charge)?

What is an action potential?

When does an action potential occur?

What is the sequence of events within one portion of the axon once an action potential has started?

What does it mean that action potentials are “all or none”?

What does the sodium/potassium pump do?

Why is the action of the sodium/potassium pump important for the proper functioning of a neuron?

Voltage-dependent membrane permeability

Describe in detail the voltage clamp method and how it works

What happens when you apply a hyperpolarizing current?

What happens when you apply a depolarizing current?

What are the differences between the early current and the late current?

Think of inward/outward, ions, shape over time, conductance etc...

What happens to the two currents when the axon is in pure water?

What happens to the two currents when TTX is applied?

What happens to the two currents when TEA is applied?

Describe how membrane potential influences permeability/conductance and ion flow.

How does an action potential move along the axon?

What is the refractory period and how does it influence the movement of the action potential?

How does axon diameter influence conduction velocity?

How does myelin influence conduction velocity?

Be familiar with some of the most common symptoms of multiple sclerosis.

What happens to myelin and axons during a MS flare up?

How might this be related to some of the symptoms of MS?

Synaptic Transmission

What are the different types of synapses found in the nervous system?

Which is most common?

What is an electrical synapse?

How does it work?

What are vesicles?

Where are they produced? (hint, could be multiple places)

Describe the process of docking and priming vesicles.

What proteins are involved in these processes?

What events take place once an action potential reaches an axon terminal?

What causes a vesicle to fuse with the presynaptic membrane?

What do neurotransmitters do once they are deposited into the synaptic cleft?

How are receptors on the post-synaptic membrane involved in the generation of post-synaptic potentials?

What are post-synaptic potentials?

How do they differ from action potentials?

How do post-synaptic potentials determine whether an action potential will fire?

What is spatial summation?

What is temporal summation?

How are NTs removed from the synapse?